ENCI335
Structural Analysis and Systems I
Tutorial 3 Solution

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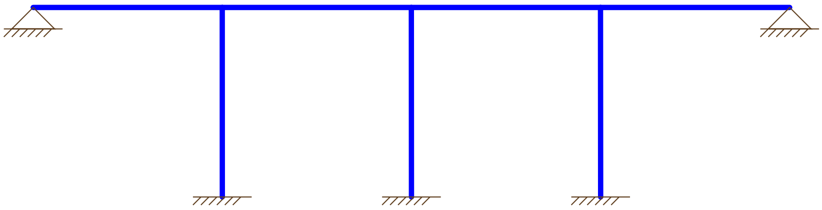
Objectives

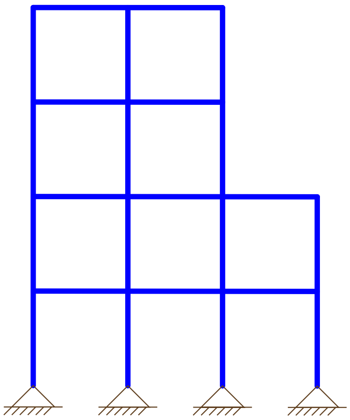
This tutorial provides an opportunity to practice the following skills:

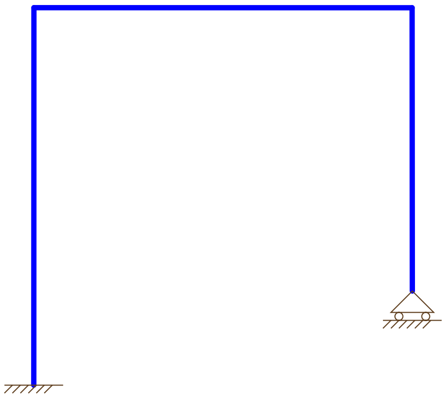
- Determination of degrees of statical indeterminacy (SI) of frames.
- Sketch the BMD, SFD, AFD, and deflected shape of frames.
- Analysis of statically determinate frames.

Question 1 Determination of SI

Determine the degree of static indeterminacy SI of the following frames.







Solution:

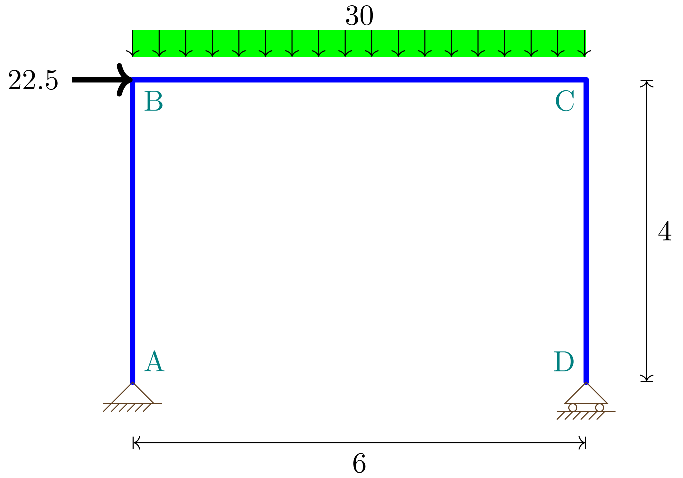
$$\text{Bridge: } n_R = 3 \times 3 + 2 \times 2 = 13, \quad n_M = 7, \quad n_J = 8, \quad SI = n_R + 3n_M - 3n_J = 10 \tag{1}$$

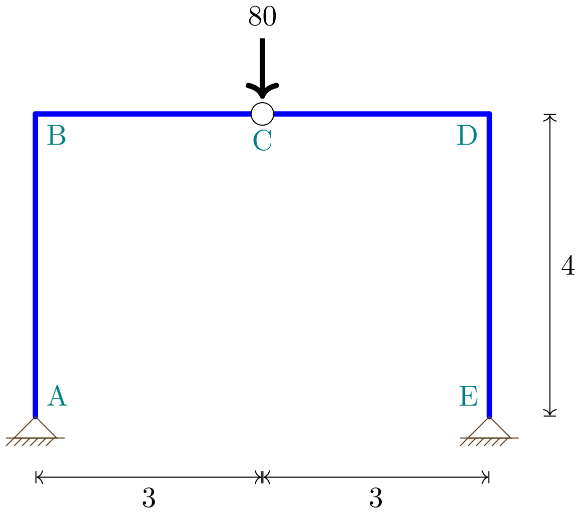
$$\text{4-storey: } n_R = 4 \times 2 = 8, \quad n_M = 24, \quad n_J = 18, \quad SI = n_R + 3n_M - 3n_J = 26 \tag{2}$$

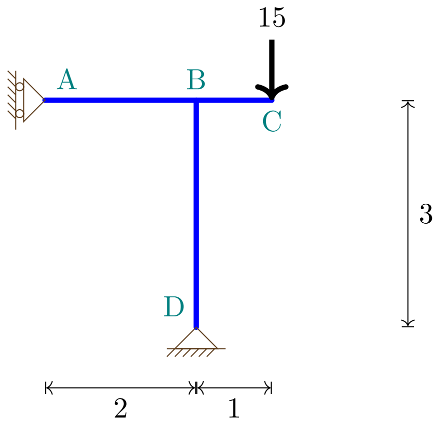
$$\text{Portal frame: } n_R = 3 + 1 = 4, \quad n_M = 3, \quad n_J = 4, \quad SI = n_R + 3n_M - 3n_J = 1 \tag{3}$$

Question 2 Deflected Shapes, BMD, SFD, and AFD

Consider the following frames, solve for the support reactions, draw their BMD, SFD, AFD, and deflected shapes.







Solution:

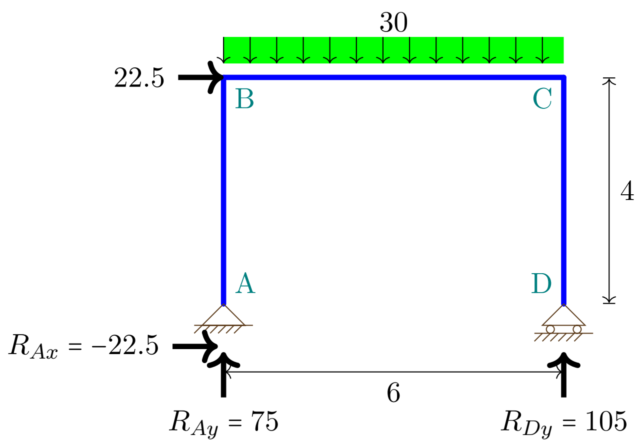
Question 2.1 Frame 1

Using the global equilibrium equations, we can obtain the support reactions as follows

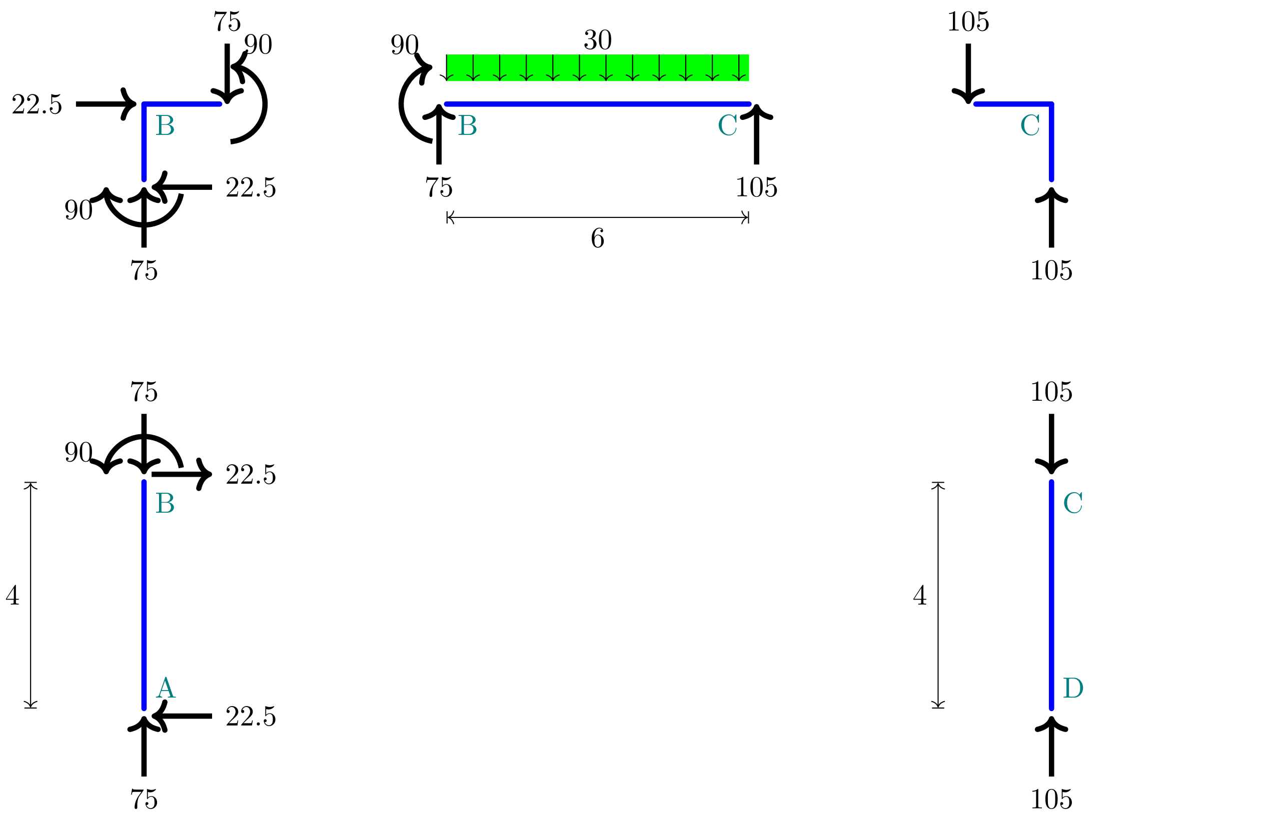
$$\sum F_X = 0 \implies R_{Ax} = -22.5 \tag{4}$$

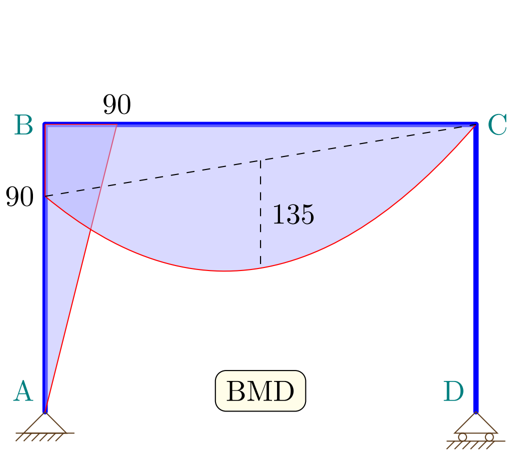
$$\sum M_Z@A = 0 \implies R_{Dy} = \frac{1}{6}(30 \times 6 \times 3 + 22.5 \times 4) = 105 \tag{5}$$

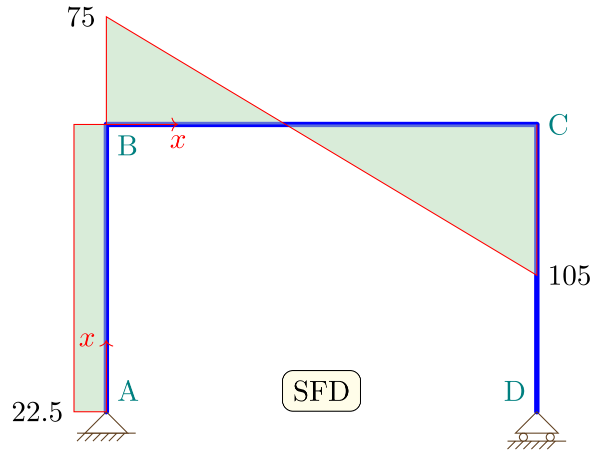
$$\sum F_Y = 0 \implies R_{Ay} = 30 \times 6 - R_{Dy} = 75 \tag{6}$$

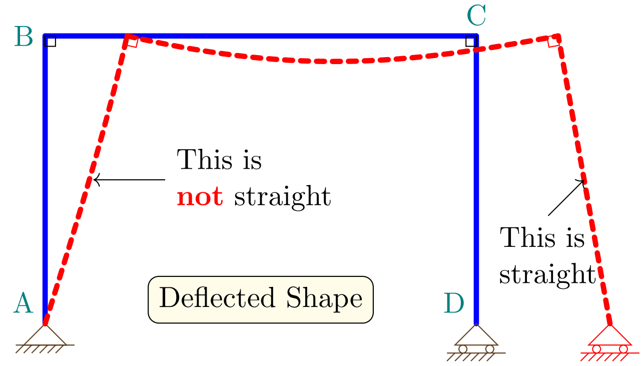


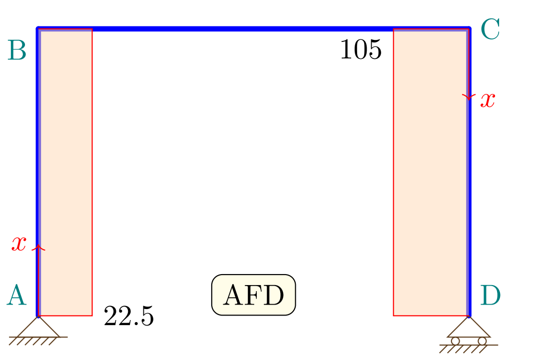
The internal forces of each member can be obtained by drawing a free body diagram, use the equilibrium equations ($\sum F_X = 0$, $\sum F_Y = 0$, $\sum M_Z = 0$) to determine unknown end forces, and use the results to calculate and sketch BMD, SFD, and AFD. Use the joint free body diagrams and the equilibrium equations to ensure a correct transfer of moments and forces between members. The free body diagrams of all the members are given below.


75
90
22.5
B
90
22.5
75
30
90
B
C
75
105
6
105
C
105
75
90
22.5
B
4
A
22.5
75
105
C
4
D
105


B
90
C
90
135
A
BMD
D

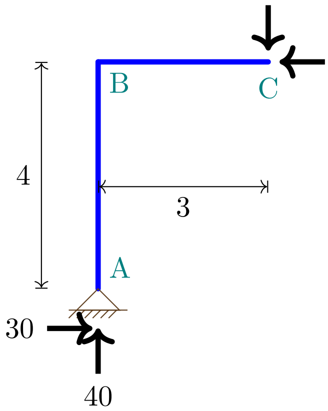

75
B
x
C
105
x
A
22.5
SFD
D

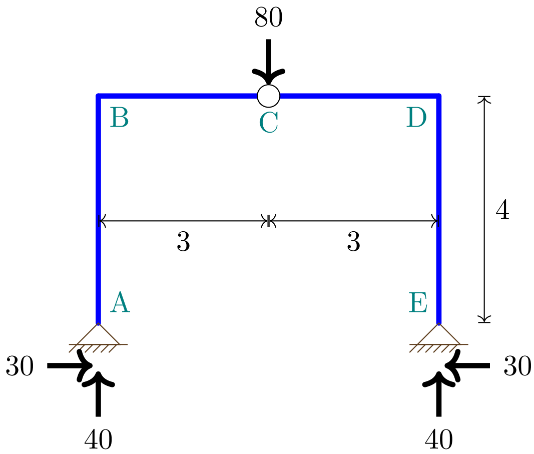

B
C
This is
not straight
This is
straight
A
Deflected Shape
D


B
105
C
x
x
A
22.5
AFD
D

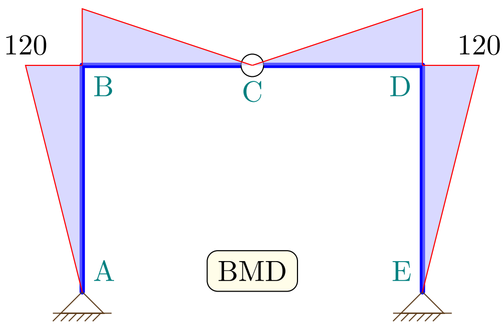
Question 2.2 Frame 2

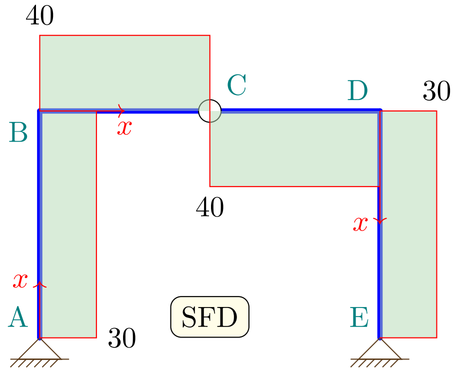
Both the frame and the load are symmetric about the middle vertical line that passes through point C. Therefore, the vertical reactions at points A and E are both 40 (upwards). To compute the horizontal reactions at both supports, we create a free body diagram for the left half structure (see below). Using the fact that the moment at the internal hinge at point C is zero, we can use the moment equilibrium equation to obtain the horizontal reaction at point A as 30 (rightwards).

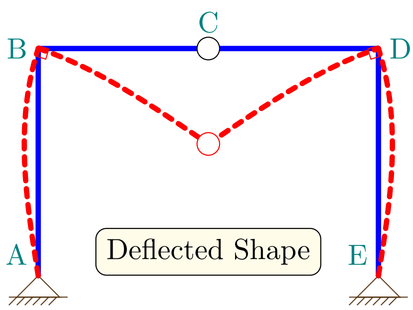


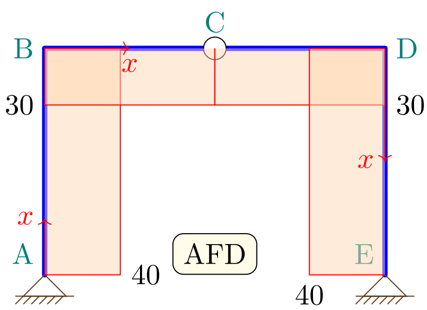


Using the information of reaction forces, the BMD, SFD, and AFD can be readily obtained as



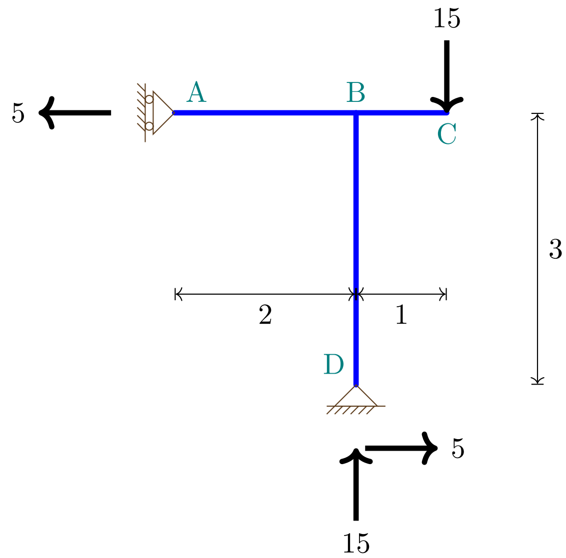






Question 2.3 Frame 3

The reactions at the supports can be obtained using the three global equilibrium equations.



Using the information of reaction forces, the BMD, SFD, and AFD can be readily obtained as